

CROSS-STAGE / NETWORK TAXONOMY AUDIT

PART 3 – FINAL TAXONOMY

FAMILY N-001: SYNTHETIC IDENTITY NETWORKS

Identity

Field	Value
Attack ID	N-001
Attack Family	Synthetic Identity Networks (SIF Networks)
Variants	Synthetic Identity Fabric (SIF); AI-generated identity clusters; Account farming networks; Synthetic identity rings
Primary Lifecycle Stage	Cross-stage / Network
Cross-stage Stages	Identity/KYC (Stage 1), Account Creation (Stage 2), Device/Session (Stage 3), Authentication (Stage 4), Payment Initiation (Stage 5), Cash-out/Mule (Stage 16)
Network Objects	Synthetic identities, accounts, devices, IP addresses, documents, digital footprints, fraud ring orchestrators
Attacker Roles	Identity generator, account opener, seasoner, orchestrator, marketplace seller
Objective	Create and operate networks of synthetic identities that pass KYC, build credit, and enable large-scale coordinated fraud

Fraud Understanding

Attack Explanation:

Attackers use GenAI to generate thousands of synthetic personas with coherent digital footprints, AI-generated identity documents, and consistent cross-platform histories. These identities are opened as accounts, "seasoned" with AI-generated legitimate-looking activity, and then activated in coordinated waves for bust-out fraud, money laundering, or sold as verified accounts on fraud marketplaces. The network structure—multiple identities under common control—is the defining characteristic.

Traditional Mechanism:

Before GenAI, synthetic identity creation was manual, using stolen PII fragments and basic forgeries. Each identity required significant effort, and networks were limited to dozens of identities. Detection was possible through document inconsistencies and manual KYC review.

GenAI Transformation:

GenAI enables industrial-scale generation of synthetic identities with flawless documents, autonomous KYC verification attempts, and APT-like persistence across thousands of accounts. Synthetic identity fraud grew eightfold in 2025, reaching 11% of global fraud.

HARD Test Result: PASS – Removing GenAI removes the ability to generate thousands of convincing synthetic identities, automate KYC, and coordinate networks at scale.

Prerequisites:

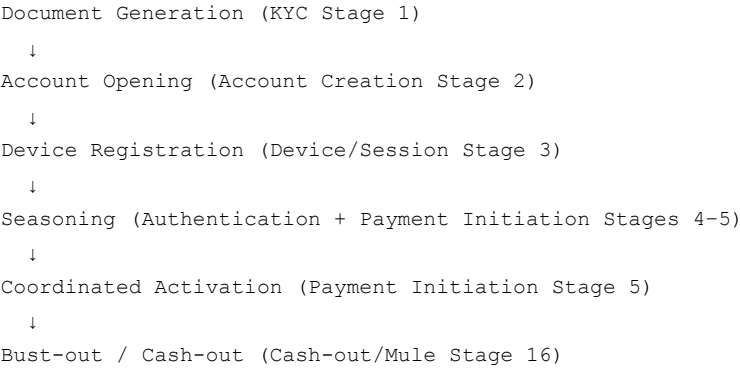
- Access to GenAI tools (LLMs, image generation, document generation)
- Breached PII datasets or ability to fabricate complete personas
- Understanding of KYC verification workflows
- Infrastructure to manage thousands of accounts

Conceptual Network Structure:

- **Nodes:** Synthetic identities, accounts, devices, IP addresses, documents
- **Edges:** Identity-account relationships, account-device relationships, shared infrastructure (IPs, devices, address clusters)
- **Roles:** Orchestrator (controls the network), Identity generator, Account opener, Seasoner (generates legitimate-looking activity), Seller (marketplace)
- **Temporal Relationships:** Identities are created in batches, seasoned over 30–365 days with AI-generated activity patterns, then activated in coordinated waves.

Cross-Stage Flow:

Identity Generation (KYC Stage 1)



Why No Single-Stage Model Is Sufficient:

Each individual identity appears legitimate to KYC; each account appears normal to Account Creation monitoring; each transaction appears normal to Payment Initiation monitoring. The fraud is in the **coordination**—the batch creation, shared infrastructure, and synchronized activation patterns. Only network-level analysis can detect these coordinated clusters.

Financial Consequence:

- Synthetic identity fraud: \$35B+ annually, 11% of global fraud, eightfold growth in 2025
- Accounts resold on fraud marketplaces
- Massive bust-out losses from credit cards and loans
- Enables downstream fraud (money laundering, mule activity)

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	False-Positive Risk
Identity	Shared identity attributes (phone, email domain, address) across "unrelated" accounts	Indicates synthetic generation or common control	Entity resolution, graph clustering, attribute similarity scoring	Large families, businesses sharing addresses
Identity	AI-generated document artifacts (pixel anomalies, metadata)	Indicates document forgery	CNN/Transformer document forensics	Poor-quality scans, legitimate documents with artifacts
Account	Batch account creation patterns (identical timing, similar attributes)	Indicates automated farming	Velocity rules, temporal clustering, attribute similarity	Legitimate business account opening campaigns
Account	Identical dormancy-to-activity patterns across accounts	Indicates orchestrated seasoning	Temporal pattern analysis, clustering	Legitimate seasonal businesses
Device	Shared device fingerprints across multiple accounts	Indicates device farm or common control	Device fingerprint correlation, graph analytics	Families sharing devices, corporate devices
Device	Superhuman form-filling speed during onboarding	Indicates bot automation	Behavioral biometrics, timing analysis	Legitimate users with accessibility tools
Transaction	Unnatural credit-building velocity	Indicates synthetic identity cultivation	Velocity rules, anomaly detection	Legitimate rapid credit-building (new affluent customers)
Transaction	Coordinated bust-out transactions across accounts	Indicates network activation	Temporal correlation, graph analytics	Legitimate seasonal spending spikes
Temporal	Identical lifecycle phases across accounts (creation → dormancy → seasoning → activation)	Indicates batch orchestration	Temporal clustering, lifecycle pattern analysis	Legitimate cohorts (students, employees)
Graph	High clustering coefficient among accounts with no legitimate relationships	Indicates synthetic network	Graph Neural Network, community detection	Corporate groups, payment aggregators
Graph	Shared infrastructure (IPs, devices, addresses) across accounts	Indicates common control	Entity resolution, graph analytics	Shared households, business parks

Simulation Intelligence

Component	Description	Method
Nodes	Synthetic identities (name, DOB, address, occupation, income, email, phone), accounts, devices, IPs	Algorithmic
Node Types	Orchestrator, identity, account, device, document, merchant, beneficiary	Algorithmic + Agentic

Component	Description	Method
Roles	Generator (creates identities), Opener (opens accounts), Seasoner (generates activity), Orchestrator (coordinates), Seller (marketplace)	Agentic
Edges	Identity-account, account-device, account-identity relationship, shared infrastructure	Algorithmic
Transactions	Seasoning transactions (salary, bills, small purchases), bust-out transactions (large transfers, maxed credit)	Algorithmic
Devices	Device fingerprints, IP addresses, geolocation	Algorithmic
Identities	Synthetic personas with coherent digital footprints (social media, email history)	Algorithmic + Agentic
Merchants	Merchant categories (MCC), amounts, timing	Algorithmic
Beneficiaries	Mule accounts, shell companies	Algorithmic
Accounts	Account histories, dormancy periods, activation timing	Algorithmic
Timestamps	Creation time, dormancy duration, seasoning patterns, activation timing	Algorithmic
Legitimate Networks	Large families, businesses, corporate groups, universities	Algorithmic
Fraudulent Networks	Synthetic identity clusters, shared infrastructure, coordinated activation	Algorithmic + Agentic
Attack Labels	Synthetic identity network, synthetic fabric, account farming	Algorithmic
Network Mutations	Attribute variation, timing variation, infrastructure rotation, activation pattern variation	Algorithmic
Cross-Stage Relationships	Identity → Account → Device → Transaction → Cash-out	Agentic

Simulation Type: Hybrid (Algorithmic for identity/account/transaction generation; Agentic for orchestration and seasoning)

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Synthetic identity fraud grew eightfold in 2025, 11% of global fraud	LexisNexis	ValidSoft	2026	Industry report	VERIFIED	DEMONSTRATED
AI-powered synthetic identities operate like advanced, networked APTs	Bureau.id	Bureau.id	2025	Industry analysis	STRONG RESEARCH	EMERGING
GenAI enables SIF—plausible, invisible networks of fake accounts	DQ India	DQ India	2025	Industry analysis	STRONG RESEARCH	EMERGING
AI fraud agents autonomously complete verification attempts	Sumsub	Sumsub	2025	Industry report	VERIFIED	DEMONSTRATED
Triad-linked ring used AI-generated facial composites to open bank accounts	Hong Kong Police	South China Morning Post	2025	Law enforcement	VERIFIED	DEMONSTRATED
Criminals use deepfake technology at scale to industrialize onboarding	ACAMS	ACAMS	2025	Industry report	VERIFIED	STRONG RESEARCH
Advanced fraud attacks surged 180% in 2025 using GenAI	Dark Reading	Dark Reading	2025	Industry report	VERIFIED	STRONG RESEARCH

FAMILY N-002: AI-ORCHESTRATED FRAUD RINGS

Identity

Field	Value
Attack ID	N-002
Attack Family	AI-Orchestrated Fraud Rings
Variants	Human-assisted fraud rings with AI content; AI-agent orchestrated rings; Multi-scam rings (romance, investment, impersonation, gambling)
Primary Lifecycle Stage	Cross-stage / Network
Cross-stage Stages	All—Identity/KYC, Account Creation, Device/Session, Authentication, Payment Initiation, Merchant, Acquirer, Gateway/Processor, Payment Rail, Cash-out/Mule, AML/Compliance

Field	Value
Network Objects	Organizers, facilitators, operators, accounts, identities, devices, merchants, shell companies, payment providers, victims
Attacker Roles	Organizers (strategic), Facilitators (payment providers, bank insiders, shell company operators), Operators (account managers, transaction executors), Content generators (AI for phishing/scams)
Objective	Execute large-scale, multi-vector fraud campaigns with coordinated nodes, maximizing financial return while evading detection through distributed operations

Fraud Understanding

Attack Explanation:

Fraud rings are organized criminal networks that coordinate multiple fraud types (credit card fraud, APP fraud, investment scams, romance scams, impersonation) using AI for content generation, phishing, and orchestration. The ring operates as a hierarchical enterprise, with specialized roles and shared infrastructure. GenAI transforms these rings from human-coordinated to AI-orchestrated, enabling industrial-scale personalized scams, autonomous campaign execution, and 4.5x higher profitability.

Traditional Mechanism:

Before GenAI, fraud rings relied on human coordination, manual phishing, and limited scalability. The Eurojust case (EUR 300M fraud) used manual shell companies, payment provider collusion, and human operators. Each campaign required significant human effort.

GenAI Transformation:

- AI agents autonomously coordinate multiple scam types simultaneously
- LLM-generated personalized content at scale (2.5M scam texts in 2 weeks)
- Autonomous campaign execution from reconnaissance to cash-out
- 4.5x profitability compared to traditional methods

HARD Test Result: PASS — Removing GenAI removes autonomous coordination, industrial-scale content generation, and multi-scam orchestration.

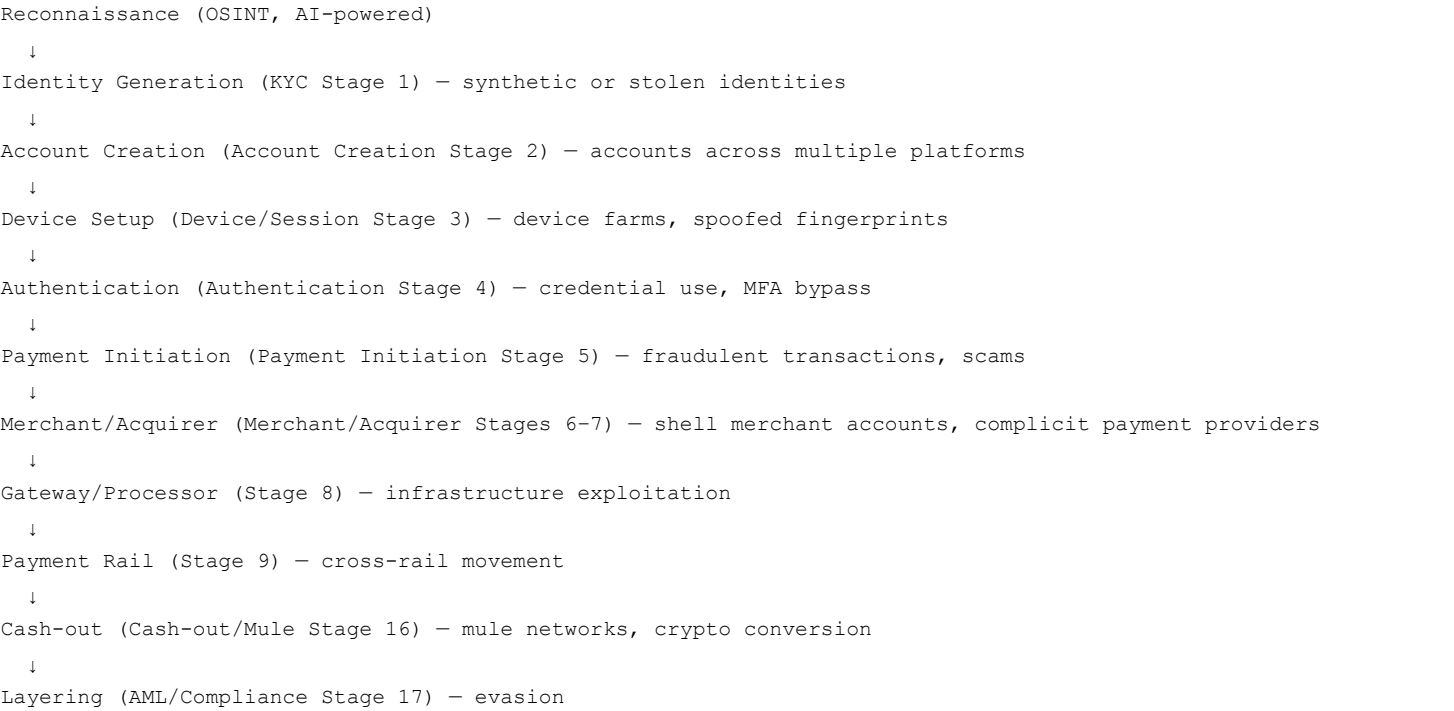
Prerequisites:

- Access to GenAI tools (LLMs, voice cloning, deepfake video)
- Infrastructure for account creation and payment processing
- Network of complicit or synthetic accounts
- Understanding of payment systems and fraud detection

Conceptual Network Structure:

- **Nodes:** Organizers, facilitators (payment providers, insiders), operators, accounts, merchants, shell companies, victims
- **Edges:** Command relationships, payment flows, shared infrastructure (IPs, devices, payment gateways), collusion relationships
- **Roles:** Organizer (strategic control), Facilitator (enables infrastructure), Operator (executes transactions), Content generator (AI for phishing/scams)
- **Temporal Relationships:** Sustained operations over months/years, with coordinated bursts of activity

Cross-Stage Flow:



Why No Single-Stage Model Is Sufficient:

Each stage appears normal in isolation—KYC passes, accounts are legitimate, transactions are within thresholds. The fraud is in the **orchestration**—the coordinated use of infrastructure, timing, and roles across stages. Only network-level analysis can detect the hierarchical structure and shared infrastructure.

Financial Consequence:

- EUR 300 million fraud (Eurojust case)
- Rs 1,000 crore cyber fraud (CBI case)
- 4.5x profitability of AI-enhanced fraud
- Billions in global losses

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	False-Positive Risk
Identity	Multiple identities linked to same facilitators or infrastructure	Indicates ring structure	Entity resolution, graph analytics	Corporate groups, payment aggregators
Account	Accounts opened with similar patterns (timing, documents)	Indicates batch creation	Temporal clustering, attribute similarity	Legitimate business account opening
Device	Shared devices, IPs, or payment gateways across accounts	Indicates common control	Device fingerprint correlation, graph analytics	Families, corporate networks
Transaction	Small, consistent transactions just below thresholds	Indicates coordinated evasion	Velocity rules, anomaly detection	Legitimate subscription billing
Transaction	Transactions across multiple accounts with similar patterns	Indicates orchestrated activity	Clustering, temporal correlation	Seasonal shopping, business cycles
Merchant	Shell merchant accounts with no physical presence	Indicates laundering front	Entity resolution, document forensics	Legitimate home-based businesses
Beneficiary	Multiple accounts sending to same mule network	Indicates fund consolidation	Graph analytics, beneficiary risk scoring	Legitimate B2B payments
Temporal	Coordinated bursts of activity across accounts	Indicates campaign execution	Temporal correlation, velocity rules	Flash sales, promotional events
Graph	High betweenness centrality for certain nodes	Indicates key facilitators	Centrality analysis, Graph Neural Network	Payment aggregators, large merchants
Graph	Community structure with dense intra-cluster and sparse inter-cluster edges	Indicates ring clusters	Community detection, graph clustering	Corporate groups, business ecosystems
Behavioral	Unnatural patterns in administrative behavior	Indicates bot or impersonation	Behavioral biometrics	Legitimate users with new devices

Simulation Intelligence

Component	Description	Method
Nodes	Organizers, facilitators, operators, accounts, merchants, shell companies, victims	Algorithmic + Agentic
Node Types	Organizer (strategic), Facilitator (infrastructure), Operator (execution), Content generator (AI)	Agentic
Roles	Strategic control, infrastructure provision, transaction execution, content generation	Agentic
Edges	Command relationships, payment flows, shared infrastructure, collusion	Algorithmic + Agentic
Transactions	Fraudulent transactions (credit card, APP, scams), layering transactions	Algorithmic
Devices	Device farms, spoofed fingerprints, shared IPs	Algorithmic
Identities	Synthetic or stolen identities, shell company identities	Algorithmic + Agentic
Merchants	Shell merchant accounts, complicit merchants	Algorithmic
Beneficiaries	Mule accounts, crypto wallets	Algorithmic
Accounts	Bank accounts, payment gateway accounts, merchant accounts	Algorithmic
Timestamps	Campaign timing, coordinated bursts, sustained operations	Algorithmic
Legitimate Networks	Corporate groups, business ecosystems, payment aggregators	Algorithmic
Fraudulent Networks	Fraud ring clusters, hierarchical structures, shared infrastructure	Algorithmic + Agentic
Attack Labels	Fraud ring, organized fraud, multi-scam campaign	Algorithmic

Component	Description	Method
Network Mutations	Role rotation, infrastructure rotation, timing variation, scam type variation	Algorithmic
Cross-Stage Relationships	All stages	Agentic

Simulation Type: Agentic (primary) with Algorithmic components

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Outsider Enterprise used Gemini for 2.5M scam texts, 1.59M fake URLs	Google/FBI	Google/FBI	2026	Law enforcement/Industry	VERIFIED	DEMONSTRATED
OpenAI disrupted ChatGPT scam network in Cambodia running multiple scam types	OpenAI	OpenAI	2026	Industry report	VERIFIED	DEMONSTRATED
Agentic AI systems autonomously plan and execute complete fraud campaigns	INTERPOL	INTERPOL	2026	Law enforcement	VERIFIED	EMERGING
Eurojust operation: EUR 300M credit card fraud, 18 arrests, 193 countries	Eurojust	Eurojust	2025	Law enforcement	VERIFIED	DEMONSTRATED
CBI chargesheeted 17 accused in Rs 1,000 cr cyber fraud with 111 shell companies	CBI	CBI	2025	Law enforcement	VERIFIED	DEMONSTRATED
73% of respondents personally affected by cyber-enabled fraud in 2025	WEF	WEF	2026	Industry report	VERIFIED	STRONG RESEARCH
AI-enhanced fraud is 4.5x more profitable	INTERPOL	INTERPOL	2026	Law enforcement	VERIFIED	EMERGING

FAMILY N-003: AI-ORCHESTRATED MULTI-STAGE FRAUD CAMPAIGNS

Identity

Field	Value
Attack ID	N-003
Attack Family	AI-Orchestrated Multi-Stage Fraud Campaigns
Variants	End-to-end autonomous campaigns; AI-planned bust-out campaigns; Agentic fraud operations
Primary Lifecycle Stage	Cross-stage / Network
Cross-stage Stages	All—Identity/KYC, Account Creation, Device/Session, Authentication, Payment Initiation, Merchant, Acquirer, Gateway/Processor, Payment Rail, Cash-out/Mule, AML/Compliance
Network Objects	All lifecycle stage entities, AI orchestrator, accounts, identities, devices, merchants, beneficiaries, mules
Attacker Roles	AI Orchestrator (autonomous planner and executor), Stage-specific agents (if multi-agent)
Objective	Execute a complete fraud operation from reconnaissance to cash-out with autonomous, coordinated execution across all stages, maximizing profit and minimizing detection

Fraud Understanding

Attack Explanation:

An autonomous AI agent (or a network of agents) plans and executes a complete fraud campaign from start to finish—identifying victims, generating synthetic identities, opening accounts, bypassing authentication, initiating fraudulent payments, layering funds through mule networks, and converting to cryptocurrency—all without human intervention. The AI adapts in real-time based on detection feedback, optimizing the campaign for maximum payout and minimum detection.

Traditional Mechanism:

Before GenAI, multi-stage fraud campaigns required human coordination across specialists (identity forgers, account openers, money movers, etc.). Each stage had manual handoffs, leading to delays, errors, and detectable patterns. Campaigns were limited in scale and required significant human effort.

GenAI Transformation:

- Autonomous planning and execution from reconnaissance to cash-out
- Real-time adaptation to detection feedback
- Seamless coordination across all lifecycle stages
- 24/7 operation, no human error or fatigue

- 4.5x profitability

HARD Test Result: PASS — Removing GenAI removes the autonomous orchestration and end-to-end execution capability.

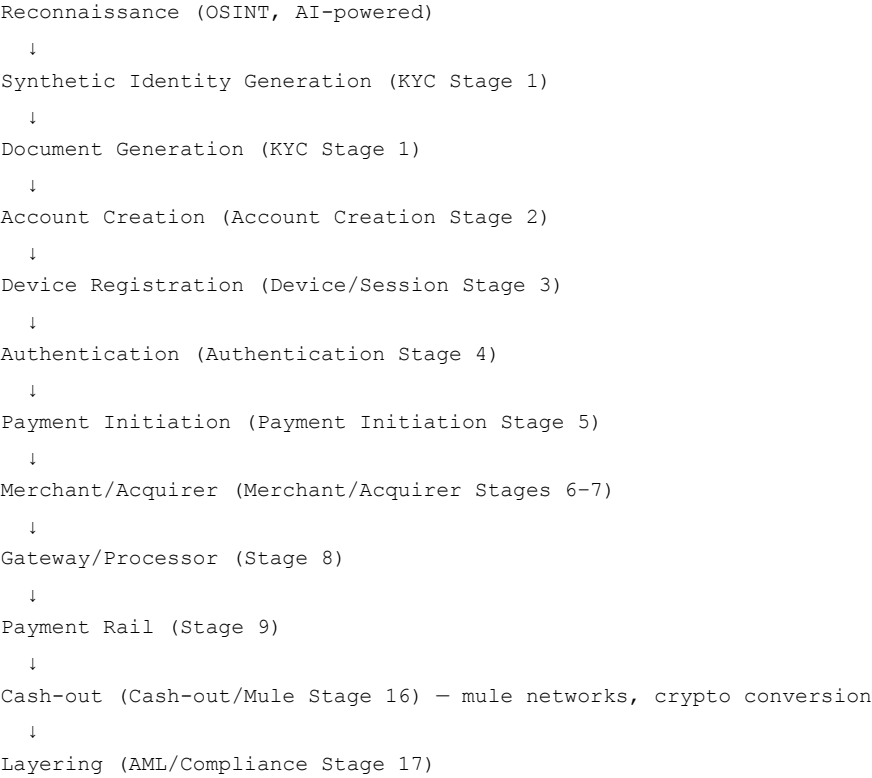
Prerequisites:

- Access to AI agent infrastructure
- Integration with multiple financial systems and payment rails
- Understanding of fraud detection systems (to evade)
- Capability to generate synthetic identities, documents, and transaction patterns

Conceptual Network Structure:

- **Nodes:** AI Orchestrator, stage-specific entities (identities, accounts, devices, merchants, mules, crypto wallets)
- **Edges:** Control flows from AI orchestrator to each stage; payment flows from fraud to cash-out
- **Roles:** AI Orchestrator (plans, executes, adapts), Stage Executors (may be sub-agents or automated scripts)
- **Temporal Relationships:** Sequential flow from reconnaissance to cash-out, with coordinated timing to maximize success

Cross-Stage Flow:



Why No Single-Stage Model Is Sufficient:

Each stage appears normal in isolation. The fraud is in the **orchestrated sequence**—the campaign is a single cohesive operation that spans all stages. Only temporal correlation across stages and graph analysis of the lifecycle can detect the campaign.

Financial Consequence:

- Complete fraud operations at scale (Rs 1,000 crore in one case)
- 4.5x profitability of AI-enhanced fraud
- Massive financial losses

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	False-Positive Risk
Identity	Batch creation of synthetic identities	Indicates campaign start	Entity resolution, temporal clustering	Legitimate business account creation
Account	Accounts opened shortly after identity creation	Indicates sequential flow	Cross-stage temporal correlation	Legitimate onboarding
Device	Unusual device patterns during onboarding	Indicates automation	Behavioral biometrics	Legitimate users with new devices

Signal Category	Observable Signal	Why It Matters	Detection Approach	False-Positive Risk
Transaction	Coordinated transactions across accounts	Indicates campaign execution	Temporal correlation, graph analytics	Flash sales, promotional events
Merchant	Shell merchant accounts used for laundering	Indicates laundering	Entity resolution, document forensics	Legitimate new businesses
Beneficiary	Funds movement to mule networks	Indicates cash-out	Graph analytics, beneficiary risk scoring	Legitimate B2B payments
Temporal	Temporal correlation across stages (identity creation → account opening → payment → cash-out)	Indicates campaign	Cross-stage temporal analysis	Legitimate business expansions
Graph	Directed chain of entities across stages	Indicates campaign structure	Graph path analysis, Graph Neural Network	Complex business transactions
Behavioral	Unnatural patterns across stages	Indicates automation	Behavioral analytics	Legitimate complex operations

Simulation Intelligence

Component	Description	Method
Nodes	AI orchestrator, identities, accounts, devices, merchants, mules, crypto wallets	Algorithmic + Agentic
Node Types	AI Orchestrator, Identity, Account, Device, Merchant, Mule, Wallet	Agentic
Roles	Planner, Executor, Adapter	Agentic
Edges	Control flows, payment flows, sequential relationships	Algorithmic + Agentic
Transactions	Fraudulent transactions, layering transactions, crypto conversions	Algorithmic
Devices	Device fingerprints, IP addresses	Algorithmic
Identities	Synthetic identities with coherent digital footprints	Algorithmic + Agentic
Merchants	Shell merchant accounts	Algorithmic
Beneficiaries	Mule accounts, crypto wallets	Algorithmic
Accounts	Bank accounts, payment gateway accounts	Algorithmic
Timestamps	Sequential timing across stages, coordinated bursts	Algorithmic
Legitimate Networks	Complex business transactions, corporate restructurings	Algorithmic
Fraudulent Networks	End-to-end campaign chains	Agentic
Attack Labels	Multi-stage campaign, end-to-end fraud, AI-orchestrated campaign	Algorithmic
Network Mutations	Stage sequence variation, timing variation, adaptation to detection	Agentic
Cross-Stage Relationships	All stages	Agentic

Simulation Type: Agentic (primary) with Algorithmic components

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Agentic AI autonomously plans and executes complete fraud campaigns	INTERPOL	INTERPOL	2026	Law enforcement	VERIFIED	EMERGING
AI-enhanced fraud is 4.5x more profitable	INTERPOL	INTERPOL	2026	Law enforcement	VERIFIED	EMERGING
CBI dismantled transnational cyber fraud network with multi-stage operation	CBI	CBI	2025	Law enforcement	VERIFIED	DEMONSTRATED
Each stage of the operation was deliberately structured to conceal identities	CBI	CBI	2025	Law enforcement	VERIFIED	DEMONSTRATED

FAMILY N-004: CROSS-RAIL / CROSS-CHAIN FRAUD NETWORKS

Identity

Field	Value
Attack ID	N-004
Attack Family	Cross-Rail / Cross-Chain Fraud Networks
Variants	Banking-to-crypto laundering; Multi-blockchain routing; Cross-chain bridge exploitation; AI-optimized cross-rail movement
Primary Lifecycle Stage	Cross-stage / Network
Cross-stage Stages	Payment Initiation (Stage 5), Payment Rail (Stage 9), Settlement (Stage 13), Cash-out/Mule (Stage 16), AML/Compliance (Stage 17)
Network Objects	Bank accounts, crypto exchanges, cross-chain bridges, blockchains, mule accounts, crypto wallets, AI routing optimizer
Attacker Roles	AI Router (optimizes cross-chain movement), Exchange operator, Bridge operator, Mule network operator, Cash-out specialist
Objective	Move illicit funds across multiple payment rails and blockchains to obscure origin, evade detection, and enable untraceable cash-out

Fraud Understanding

Attack Explanation:

Attackers use GenAI to autonomously move funds across traditional payment systems and multiple blockchains, optimizing routing through cross-chain bridges and decentralized exchanges to destroy source traceability. AI agents split funds, select optimal bridging routes, and execute swaps on decentralized platforms, enabling complete cash-out before detection systems can react. The network spans multiple rails (banking, UPI, cards, crypto) and dozens of blockchains.

Traditional Mechanism:

Before GenAI, criminals moved funds manually through crypto exchanges, often leaving detectable patterns. Cross-chain movement was limited and slow, with manual selection of bridges and exchanges. Detection was possible through tracing on single blockchains.

GenAI Transformation:

- Autonomous cross-chain routing optimization
- Real-time adaptation to detection and gas fees
- Industrial-scale cross-chain movement (6+ blockchains in minutes)
- 63% of illicit on-chain volume uses stablecoins

HARD Test Result: PARTIAL — Cross-rail movement exists without GenAI, but GenAI enables autonomous, optimized, high-speed cross-chain routing at scale that materially changes the attack.

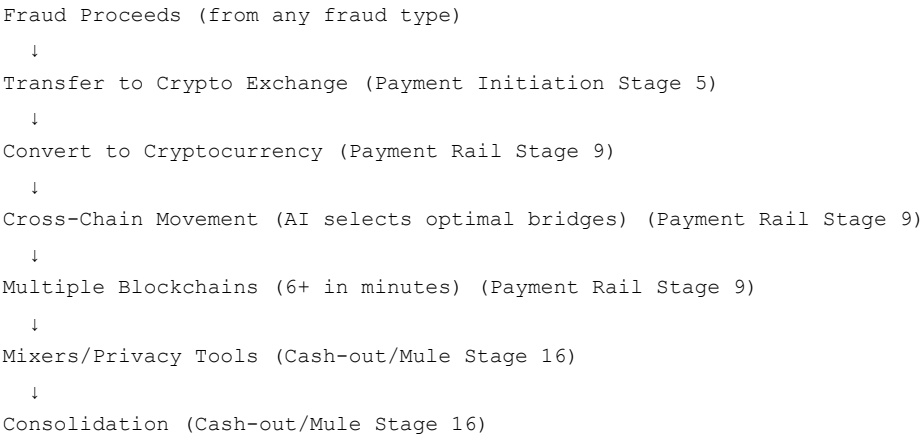
Prerequisites:

- Access to AI agent infrastructure
- Crypto exchange accounts (weak KYC preferred)
- Knowledge of cross-chain bridges and routing
- Access to mule accounts with funds

Conceptual Network Structure:

- **Nodes:** Bank accounts, crypto exchanges, blockchains, cross-chain bridges, mule accounts, crypto wallets, AI router
- **Edges:** Fund flows between accounts, exchanges, bridges, blockchains
- **Roles:** AI Router (optimizes movement), Exchange operator, Bridge operator, Mule network operator, Cash-out specialist
- **Temporal Relationships:** Rapid sequential movement (seconds/minutes), cross-chain swaps in milliseconds

Cross-Stage Flow:



↓
Fiat Conversion (Cash-out/Mule Stage 16)
↓
Criminal Organization

Why No Single-Stage Model Is Sufficient:

Each rail is monitored separately—banking, crypto exchanges, blockchains have siloed detection. The fraud is in the **cross-system movement**—the orchestrated transition across rails. Only cross-rail graph analysis can detect the pattern.

Financial Consequence:

- Industrial-scale money laundering
- Billions of dollars moved through crypto
- 63% of illicit on-chain volume in stablecoins
- AI-enabled fraud schemes are 4.5x more profitable

Detection Intelligence

Signal Category	Observable Signal	Why It Matters	Detection Approach	False-Positive Risk
Account	Funds moving from bank accounts to crypto exchanges	Indicates potential laundering	Transaction monitoring, exchange detection	Legitimate crypto investments
Transaction	Rapid conversion of large amounts to stablecoins	Indicates cash-out preparation	Anomaly detection, velocity rules	Legitimate large crypto purchases
Transaction	Multiple cross-chain bridge transactions in short sequence	Indicates cross-chain laundering	Cross-chain analytics, bridge monitoring	Legitimate DeFi activity
Transaction	Use of privacy tools (mixers, tumblers)	Indicates obfuscation	Blockchain analytics, mixers detection	Privacy-conscious users
Temporal	Conversion and cross-chain movement within seconds/minutes	Indicates autonomous AI execution	Temporal correlation, velocity rules	Legitimate high-frequency trading
Graph	Multi-hop graph from banking to multiple blockchains	Indicates cross-rail laundering	Graph Neural Network, cross-system analysis	Legitimate complex financial flows
Behavioral	Optimized routing patterns (low fees, fast confirmation)	Indicates AI optimization	Behavioral analytics, optimization detection	Legitimate DeFi optimization

Simulation Intelligence

Component	Description	Method
Nodes	Bank accounts, crypto exchanges, blockchains, cross-chain bridges, mule accounts, crypto wallets	Algorithmic
Node Types	Bank account, Exchange, Blockchain, Bridge, Mule account, Wallet	Algorithmic
Roles	AI Router, Exchange operator, Bridge operator, Mule operator, Cash-out specialist	Agentic
Edges	Fund flows between nodes, bridge connections, exchange conversions	Algorithmic + Agentic
Transactions	Banking transfers, crypto conversions, cross-chain swaps	Algorithmic
Devices	Device fingerprints for account access	Algorithmic
Identities	Synthetic or stolen identities for accounts	Algorithmic
Merchants	Not primary	N/A
Beneficiaries	Crypto wallets, exchange accounts	Algorithmic
Accounts	Bank accounts, exchange accounts	Algorithmic
Timestamps	Rapid sequential movement, cross-chain timing	Algorithmic
Legitimate Networks	Legitimate crypto investments, DeFi activity, cross-border payments	Algorithmic
Fraudulent Networks	Cross-rail laundering, cross-chain movement	Algorithmic + Agentic
Attack Labels	Cross-rail laundering, cross-chain routing, crypto cash-out	Algorithmic
Network Mutations	Bridge selection variation, timing variation, blockchain diversity	Algorithmic
Cross-Stage Relationships	Payment Initiation → Payment Rail → Cash-out	Agentic

Simulation Type: Hybrid (Agentic for AI routing optimization; Algorithmic for transaction generation)

Evidence

Claim	Source	Publisher	Date	Evidence Type	Confidence	Maturity
Generative AI automates cross-chain crypto laundering across dozens of blockchains	FINX Insights	FINX Insights	2026	Industry analysis	VERIFIED	EMERGING
Agentic AI autonomously plans and executes complete fraud campaigns	INTERPOL	INTERPOL	2026	Law enforcement	VERIFIED	EMERGING
63% of illicit on-chain volume uses stablecoins	Chainalysis	Chainalysis	2025	Industry report	VERIFIED	STRONG RESEARCH
Fraudsters drain accounts, transfer to mules, convert to stablecoins	Chainalysis	Chainalysis	2025	Industry report	VERIFIED	STRONG RESEARCH
Criminal groups use cross-chain bridges to move assets quickly and hide traces	Elliptic	Elliptic	2025	Industry report	VERIFIED	STRONG RESEARCH
Online scam losses reached \$442 billion globally in 2025	Global Anti-Scam Alliance	Global Anti-Scam Alliance	2026	Industry report	VERIFIED	STRONG RESEARCH

PART 4 – COVERAGE AUDIT

Network Surface	Considered?	Covered By	Separate Family?	Decision	Reason
Fraud Rings	✓	N-002	Yes	KEEP	Distinct hierarchical coordination across multiple fraud types; GenAI load-bearing
Synthetic Identity Networks	✓	N-001	Yes	KEEP	Coordinated cluster of synthetic identities; not covered by single-identity KYC/Account Creation
Mule Networks	✓	Cash-out/Mule (CM-003)	No	MOVE	Already covered in Cash-out/Mule taxonomy
Device-Account Networks	✓	Device/Session (EFF-001, BOT-001)	No	MOVE	Already covered in Device/Session taxonomy
Merchant-Customer Collusion	✓	Merchant (MCH-002, MCH-004)	No	MOVE	Already covered in Merchant taxonomy
Distributed Account Networks	✓	N/A	No	THEORETICAL	Plausible but insufficient evidence of GenAI transformation
Cross-Rail Networks	✓	N-004	Yes	KEEP	Distinct movement across payment rails and blockchains; GenAI strongly partial
Multi-Stage Campaigns	✓	N-003	Yes	KEEP	End-to-end AI-orchestrated campaigns; GenAI load-bearing
Coordinated Transaction Networks	✓	N/A	No	THEORETICAL	Plausible but insufficient evidence
Shared Infrastructure	✓	N-001, N-002	No	MERGED	Covered within Synthetic Identity Networks and Fraud Rings
Temporal Coordination	✓	N-002, N-003	No	MERGED	Covered within Fraud Rings and Multi-Stage Campaigns
Adaptive Network Behavior	✓	N-002, N-003, N-004	No	MERGED	Covered across families
AI-Coordinated Fraud Networks	✓	N-002, N-003	No	MERGED	Covered within Fraud Rings and Multi-Stage Campaigns
Synthetic Personas Across Multiple Nodes	✓	N-001	No	MERGED	Covered within Synthetic Identity Networks
Agentic Coordination	✓	N-002, N-003	No	MERGED	Covered within Fraud Rings and Multi-Stage Campaigns

PART 5 – PRODUCT MAPPING

RED TEAM – Synthetic Fraud Network Generation

The system generates synthetic fraud networks by composing algorithmic and agentic components:

Network Generation:

- Algorithmic:** Generate node attributes (identities, accounts, devices, merchants, beneficiaries) with realistic distributions, relationships, and temporal patterns.
- Agentic:** AI agents orchestrate network behavior—coordinating transactions, adapting to detection, and evolving network structure.
- Hybrid:** Combine algorithmic parameterization with agentic orchestration for complex, realistic networks.

Network Parameters That Can Mutate:

- Network size (number of nodes)
- Node attribute variation (diversity of identities, account types)
- Graph topology (star, mesh, chain, hierarchical, bipartite)
- Temporal patterns (synchronized, phase-shifted, anti-correlated)
- Infrastructure sharing (IPs, devices, addresses)
- Transaction patterns (amounts, timing, beneficiary diversity)
- Detection evasion techniques (staying below thresholds, pattern-free movement)

Legitimate Networks as Controls:

- Generate legitimate networks (families, businesses, payment aggregators, marketplaces) with similar structural properties but different behavioral profiles.
- Use these as controls to train the defender to distinguish legitimate from fraudulent networks.

GENERATE – Classification

Component	Classification	Description
Synthetic identities	Algorithmic	Parameterized personas with consistent attributes
Synthetic documents	Algorithmic	AI-generated documents with varying quality
Account creation	Algorithmic + Agentic	Automated account opening with behavioral simulation
Device fingerprints	Algorithmic	Realistic device profiles
Transaction sequences	Algorithmic	Parameterized transaction patterns (seasoning, bust-out, layering)
Network orchestration	Agentic	AI agents coordinating across accounts, stages, and entities
Cross-rail movement	Hybrid	Algorithmic routing + Agentic optimization
Adaptive behavior	Agentic	Real-time adaptation to detection feedback
Mutation	Algorithmic	Variation of network parameters
Attack labels	Algorithmic	Classification of network type and fraud stage

DEFEND – Detection Strategy

Combine multiple models for network-level detection:

1. Transaction-Level Model:

- Features: Amount, merchant, time, location, velocity, beneficiary novelty
- Model: ML anomaly detection (Isolation Forest, Autoencoder)
- Output: Per-transaction risk score

2. Account-Level Behavioral Model:

- Features: Account history, lifecycle stage, dormancy patterns, credit-building velocity
- Model: Sequence modeling (LSTM/Transformer), behavioral analytics
- Output: Per-account risk score

3. Graph/Network Model:

- Features: Node degree, centrality, clustering coefficient, shared infrastructure, community membership
- Model: Graph Neural Network (GNN), community detection, entity resolution
- Output: Node-level and edge-level risk scores, suspicious clusters

4. Temporal Model:

- Features: Temporal correlation across stages, synchronized activity, phase-shifted patterns

- Model: Temporal graph analysis, cross-correlation, lifecycle pattern detection
- Output: Temporal anomaly score, coordinated activity detection

5. **Cross-Stage Correlation:**

- Features: Identity → Account → Device → Transaction → Cash-out chains
- Model: Path analysis, graph traversal, lifecycle correlation
- Output: Campaign-level risk score

Integration:

- Combine outputs into an ensemble risk score.
- Flag suspicious clusters, relationships, and entities for enhanced review.
- Escalate to human review for high-risk networks.
- Apply hold/restrict actions where appropriate.

FEEDBACK – Closing the Loop

If the Defender Detects Individual Transactions but Misses the Network:

1. **Identify the Blind Spot:**

- Monitor detection performance on network-level metrics (cluster detection, graph-based precision/recall).
- Identify network patterns with high false negative rates (e.g., certain graph topologies, temporal patterns).

2. **Generate Harder Network Variants:**

- Mutate network parameters to exploit the blind spot:
 - Change topology (e.g., from star to mesh)
 - Vary temporal coordination (e.g., from synchronized to phase-shifted)
 - Alter node attributes (e.g., increase identity diversity)
 - Introduce noise (e.g., add legitimate nodes to dilute signal)

3. **Retrain/Retest:**

- Retrain the graph model on the harder variants.
- Test if detection improves.
- Repeat until performance on network-level metrics is satisfactory.

4. **Measure Improvement:**

- Track precision, recall, and F1 for network-level detection (cluster detection, campaign identification).
- Measure reduction in false negatives for coordinated fraud.

PART 6 – FINAL SUMMARY

Table 1 – FINAL NETWORK TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Cross-stage Stages	Reason for Distinctness
N-001	Synthetic Identity Networks	Synthetic Identity Fabric, AI-generated identity clusters, Account farming networks	PASS	Hybrid	Cross-stage/Network	KYC, Account Creation, Device/Session, Authentication, Payment Initiation, Cash-out/Mule	Coordinated cluster of synthetic identities under common control; not covered by single-identity KYC/Account Creation
N-002	AI-Orchestrated Fraud Rings	Human-assisted fraud rings with AI content; AI-agent orchestrated rings; Multi-scam rings	PASS	Agentic	Cross-stage/Network	All stages	Hierarchical coordination across multiple fraud types; GenAI enables autonomous orchestration and industrial-scale content
N-003	AI-Orchestrated Multi-Stage Fraud Campaigns	End-to-end autonomous campaigns; AI-planned bust-out campaigns; Agentic fraud operations	PASS	Agentic	Cross-stage/Network	All stages	End-to-end AI-orchestrated campaign spanning all lifecycle stages; autonomous planning and execution

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Cross-stage Stages	Reason for Distinctness
N-004	Cross-Rail / Cross-Chain Fraud Networks	Banking-to-crypto laundering; Multi-blockchain routing; Cross-chain bridge exploitation	PARTIAL	Hybrid	Cross-stage/Network	Payment Initiation, Payment Rail, Settlement, Cash-out/Mule, AML/Compliance	Movement across payment rails and blockchains; GenAI enables autonomous, optimized cross-chain routing

Table 2 – ATTACK × NETWORK SIGNAL

Attack Family	Identity	Account	Device	Transaction	Merchant	Beneficiary	Temporal	Graph	Behavioral
N-001: Synthetic Identity Networks	Shared attributes, AI-generated documents, batch creation	Batch creation, identical dormancy patterns	Shared fingerprints, superhuman speed	Unnatural credit-building, coordinated bust-out	Not primary	Not primary	Identical lifecycle phases, batch timing	High clustering, shared infrastructure	Unnatural form-filling, automation patterns
N-002: Fraud Rings	Multiple identities linked to facilitators	Shared opening patterns, shell accounts	Shared infrastructure (IPs, devices)	Coordinated transactions, small thresholds	Shell merchant accounts, complicit merchants	Mule networks, crypto wallets	Coordinated bursts, sustained operations	High betweenness centrality, community structure	Unnatural administrative behavior, bot patterns
N-003: Multi-Stage Campaigns	Batch creation at campaign start	Accounts opened shortly after identities	Device patterns during onboarding	Coordinated transactions across stages	Shell merchants used for laundering	Mule networks, crypto wallets	Cross-stage temporal correlation, coordinated bursts	Directed chain across stages	Automation patterns across stages
N-004: Cross-Rail Networks	Not primary	Funds moving to crypto exchanges	Not primary	Rapid conversion to stablecoins, cross-chain swaps	Not primary	Crypto wallets, exchange accounts	Rapid sequential movement, cross-chain timing	Multi-hop graph from banking to multiple blockchains	Optimization patterns (low fees, fast confirmation)

Table 3 – ATTACK × SIMULATION

Attack Family	Nodes	Edges	Transactions	Devices	Accounts	Merchants	Timeline	Network Mutations	Agentic/Algorithmic/Hy
N-001: Synthetic Identity Networks	Synthetic identities, accounts, devices, IPs	Identity-account, account-device, shared infrastructure	Seasoning transactions, bust-out transactions	Device fingerprints, IP addresses	Account histories, dormancy, activation	Merchant categories	Creation → dormancy → seasoning → activation	Attribute variation, timing variation, infrastructure rotation	Hybrid
N-002: Fraud Rings	Organizers, facilitators, operators, accounts, merchants	Command, payment flows, shared infrastructure	Fraudulent transactions, layering transactions	Device farms, spoofed fingerprints	Bank accounts, payment gateway accounts	Shell merchant accounts	Sustained operations, coordinated bursts	Role rotation, infrastructure rotation, timing variation	Agentic
N-003: Multi-Stage Campaigns	AI orchestrator, identities, accounts, devices, merchants, mules	Control flows, payment flows, sequential relationships	Fraudulent transactions, crypto conversions	Device fingerprints, IP addresses	Bank accounts, payment gateway accounts	Shell merchant accounts	Sequential flow, coordinated bursts	Stage sequence variation, timing variation, adaptation	Agentic
N-004: Cross-Rail Networks	Bank accounts, crypto exchanges, blockchains, bridges	Fund flows, bridge connections, exchange conversions	Banking transfers, crypto conversions, cross-chain swaps	Not primary	Bank accounts, exchange accounts	Not primary	Rapid sequential movement, cross-chain timing	Bridge selection variation, timing variation, blockchain diversity	Hybrid

Table 4 – TAXONOMY DECISIONS

Candidate	Decision	Reason
Synthetic Identity Networks	KEEP	Distinct network mechanism; GenAI load-bearing; not covered by single-identity KYC/Account Creation

Candidate	Decision	Reason
Fraud Rings	KEEP	Distinct network mechanism; GenAI load-bearing; not covered elsewhere
Multi-Stage Campaigns	KEEP	Distinct network mechanism; GenAI load-bearing; not covered elsewhere
Cross-Rail Networks	KEEP	Distinct network mechanism; GenAI strongly partial; not covered elsewhere
Mule Networks	MOVE → Cash-out/Mule	Already covered by Cash-out/Mule (CM-003)
Device Farms / Multi-Accounting	MOVE → Device/Session	Already covered by Device/Session (EFF-001, BOT-001)
Merchant-Customer Collusion	MOVE → Merchant	Already covered by Merchant (MCH-002, MCH-004)
Account Farming Networks	MERGE → N-001	Subset of Synthetic Identity Networks
Insider-Facilitated Networks	MOVE → Employee/Insider	GenAI incidental (FAIL)
Distributed Account Networks	THEORETICAL	Insufficient evidence of GenAI transformation
Coordinated Transaction Networks	THEORETICAL	Insufficient evidence of GenAI transformation

Final Metrics

Metric	Value
Total final network attack families	4
Total variants	12+ (across 4 families)
Important candidates merged	Account Farming Networks → N-001
Important candidates moved	Mule Networks → Cash-out/Mule; Device Farms → Device/Session; Merchant-Customer Collusion → Merchant; Insider-Facilitated Networks → Employee/Insider
Important candidates dropped	None (moved or theoretical)
Theoretical/future threats	Distributed Account Networks; Coordinated Transaction Networks
Evidence gaps	Real-world frequency of AI-orchestrated cross-chain laundering; Specific detection evasion effectiveness against production graph models; Impact of emerging GenAI architectures on network-level fraud
Network surfaces covered	Fraud rings; Synthetic identity networks; Multi-stage campaigns; Cross-rail/cross-chain networks; Shared infrastructure; Temporal coordination; Adaptive network behavior; AI-coordinated fraud; Synthetic personas across multiple nodes; Agentic coordination
Important surfaces not represented separately	Mule networks (covered in Cash-out/Mule); Device farms (covered in Device/Session); Merchant-customer collusion (covered in Merchant); Insider-facilitated networks (covered in Employee/Insider)
Existing lifecycle stages that absorb moved attacks	Cash-out/Mule (CM-003); Device/Session (EFF-001, BOT-001); Merchant (MCH-002, MCH-004); Employee/Insider

UNIFYING THEME

The Cross-stage / Network taxonomy captures the **orchestrated, coordinated, and distributed nature** of modern GenAI-enabled fraud. The unifying theme is the **shift from isolated attacks to interconnected fraud networks**:

- **Before GenAI:** Fraud was largely point-in-time, per-entity, and human-coordinated. Networks were limited in scale and detectable through manual analysis of shared infrastructure or patterns.
- **With GenAI:** Fraud networks are industrial-scale, autonomous, and adaptive. AI agents generate synthetic identities at scale, orchestrate multi-stage campaigns, coordinate fraud rings, and optimize cross-rail movement. The relationships between entities—not the entities themselves—are the primary attack surface.

The Core Vulnerability:

The payment ecosystem is a network. Fraud detection has traditionally been point-in-time and per-entity. GenAI enables fraudsters to operate at the **network level**, where individual transactions and accounts appear legitimate, but the **coordinated whole** is fraudulent.

The Attack Surface Has Shifted From:

"Can we make this transaction look legitimate?" → Transaction-level evasion

"Can we make this account look legitimate?" → Account-level evasion

To:

"Can we generate an entire network of relationships that the system accepts as legitimate?"

Fraudsters no longer rely on isolated attacks. They use GenAI to generate:

- **Synthetic identity networks** that pass KYC as individuals but are coordinated clusters
- **Fraud rings** that orchestrate multiple scams simultaneously
- **Multi-stage campaigns** that autonomously plan and execute complete fraud operations
- **Cross-rail networks** that move funds across payment systems and blockchains

This represents a fundamental change in the threat model: **the fraud network itself is the attack, and GenAI makes it indistinguishable from legitimate economic networks**. Detection must move from transactions and accounts to graph intelligence, temporal correlation, and cross-stage analysis.

END OF FINAL CROSS-STAGE / NETWORK TAXONOMY AUDIT + MASTER FRAUD INTELLIGENCE DOCUMENT